

The Geometry of Consolidation

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Abstract

The question. When a memory system replaces n cluster members with $m < n$ representatives, what geometry decides whether those representatives still recover the members by retrieval?

The law. We prove a *Consolidation–Interference Duality*: for any consolidator, whenever the retrieval cap half-angle $\theta' = 1 - \theta$ is below the mean within-cluster cosine distance $\bar{d}$, the identity error obeys $\varepsilon_{\text{id}} \geq 1 - c_1 m (\theta' / \bar{d})^{d_{\text{eff}}/2}$, where c_1 is an absolute constant and d_{eff} is the local effective (participation-ratio) dimension. The law separates a *tight regime* ($\bar{d} < \theta'$, identity cheap) from a *spread regime* ($\bar{d} \geq \theta'$, identity expensive).

The surprising consequence. Every real-text corpus we measure sits in the tight regime the law declares safe, and the cheapest consolidator there is the centroid. Across six corpora and six sentence encoders, the *fixed centroid* ties or beats every adaptive operator, including the per-cluster router we built to probe the law. Centroid dominance is not a heuristic; it is what the law predicts on real text.

Evidence. We validate the regime boundary on a 400-cell synthetic grid (16,000 trials): tight-regime cells achieve mean cap-error $\leq 0.5\%$; spread-regime cells diverge to 30–74% with an order predicted by d_{eff} . We calibrate the bound empirically: the tight-regime $c_1^{95} \approx 0.05$ on 12,400 cells — within an order of magnitude of the ideal Berry–Esseen constant; in the spread regime c_1 grows unbounded as $\theta \rightarrow 1$. On retrieval-augmented Llama-3.1-70B-Instruct QA, centroid *hurts* NQ by 4.2 EM, is neutral on HotpotQA, and *wins* PopQA by 8.4 EM over GAC — matching the cap-coverage prediction of which operator the law prefers per corpus. We treat Geometry-Aware Consolidation (GAC) not as a product but as the instrument that made the measurement.

Scope. Embedding consolidation in contrastive unit-norm text spaces under cosine-threshold retrieval. The isotropic cap-volume bound is shape-correct in the tight regime and loose-to-vacuous in the extreme-coherence corner of the spread regime; an anisotropic refinement is the central open theoretical problem. All 17,813 experimental cells are open-source under MIT license.

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Introduction

Memory systems that summarize old traces into representatives – cognitive consolidation^{1,2}, retrieval-augmented generation with vector-store compression³, replay buffers in continual learning⁴ – all rest on an implicit hope: the important structure is low-dimensional, so a few representatives can stand in for many traces without losing identity. The previous two papers in this trilogy challenged that hope. The first showed that production embedding spaces concentrate $\sim 99\%$ of their variance in $d_{\text{eff}} \approx 16$ effective dimensions⁵; the second showed that *any* kernel-threshold memory with finite d_{eff} forgets by a lower bound that no amount of storage can evade⁶. Consolidation looks like a way out: rather than raising capacity, we lower bookkeeping. In this paper we show that consolidation shares the same geometric floor. The geometry that forces forgetting also forces identity collapse under compression, by an analogue of the same argument.

Beyond the theoretical statement, a conference-level evaluation requires that we benchmark consolidation operators against the dominant vector-store quantization family in production systems – product quantization⁷, optimized product quantization⁸, locality-sensitive hashing⁹, PCA with int-8 coding, and HNSW with pruning. We test six sentence-encoder families (BGE-base/large¹⁰, MPNet¹¹, MiniLM¹², E5-large¹³, Nomic¹⁴) to separate theorem-predicted universality from encoder-specific artefacts, and we measure identity under paraphrase queries to distinguish the bound’s geometric regime from a purely memorization regime.

Our contributions are:

- (C1) **Consolidation–Interference Duality** – a theorem relating identity retrieval after compression to the same spectral quantity that governs retrieval under noise, with a full proof via a cap-volume lemma and union bound (§ *The duality theorem*, Methods).
- (C2) **Identity–Coverage decomposition** – two orthogonal axes of consolidation quality (identity of stored items vs. coverage of queries) that each obey their own regime law.
- (C3) **Geometry-Aware Consolidation (GAC)** – a per-cluster routing algorithm that Pareto-dominates medoid on every synthetic $(d_{\text{eff}}, \theta, \bar{d})$ cell.
- (C4) **Ablation-level honest findings** – on five real-text corpora the centroid picker dominates full GAC routing by 1–6 identity-accuracy points; adaptive residual-direction routing helps only on the synthetic DRM corpus.
- (C5) **Encoder and compression-family universality** – GAC’s strategy ranking reproduces across six encoders and Pareto-trades against five learned-quantization baselines at matched bytes/vector.
- (C6) **Paraphrase robustness** – at embedding-cosine 0.92 (realistic paraphrase), identity accuracy drops by < 0.02 across all strategies, refuting a pure-memorization explanation.

Results

The duality theorem

Let $\mathcal{C} = \{x_1, \dots, x_n\} \subset S^{d-1}$ be a cluster on the unit sphere with mean cosine distance $\bar{d} = \frac{1}{\binom{n}{2}} \sum_{i < j} (1 - \langle x_i, x_j \rangle)$, and let $\mathcal{R} = \{r_1, \dots, r_m\} \subset S^{d-1}$ be any set of $m \leq n$ representatives produced by a consolidator. Given a retrieval threshold $\theta \in (0, 1)$, define the identity-retrieval error $\varepsilon_{\text{id}}(\mathcal{C}, \mathcal{R}) = \frac{1}{n} |\{i : \max_j \langle x_i, r_j \rangle < \theta\}|$.

Theorem 1 (Consolidation–Interference Duality). *For every consolidator and every $\mathcal{C}$ with local effective dimension d_{eff} and mean within-cluster distance $\bar{d}$ satisfying $\theta' < \bar{d}$,*

$$\varepsilon_{\text{id}}(\mathcal{C}, \mathcal{R}) \geq 1 - c_1 m \left(\frac{\theta'}{\bar{d}} \right)^{d_{\text{eff}}/2}, \quad \theta' := 1 - \theta,$$

where c_1 is an absolute constant independent of m, n and the choice of consolidator (proof in Methods).

Corollary 2 (Compression lower bound). *For any $\varepsilon^* \in (0, 1)$, achieving $\varepsilon_{\text{id}} \leq \varepsilon^*$ requires $m \geq c_1^{-1}(1 - \varepsilon^*)(\bar{d}/\theta')^{d_{\text{eff}}/2}$: the minimum representative budget grows exponentially in the local effective dimension.*

Corollary 3 (Isotropic degeneracy). *On isotropic clusters, importance-weighted consolidation converges to the uniform centroid at rate $O(1/n)$; it therefore inherits the centroid’s bound with no improvement.*

Empirical validation of the bound (Experiment E1)

We generate a 400-cell grid over $d_{\text{eff}} \in \{4, 8, 16, 32, 64\}$, $\theta \in \{0.70, 0.80, 0.90, 0.95\}$, $\bar{d} \in \{0.05, 0.10, \dots, 1.00\}$, sampling $n = 1500$ points per cluster with 10 random seeds per cell (16,000 trials total). For each cell we run four strategies (centroid, medoid, 50% selective pruning, GAC) and measure the native quantity of the theorem – the cap-coverage error $\varepsilon_{\text{cap}} := \Pr_{x \sim \mathcal{C}}[\max_j \langle x, r_j \rangle < \theta]$.

Regime separation. The 400 cells split into tight ($\bar{d} < \theta'$) and spread ($\bar{d} \geq \theta'$) halves. In the tight regime every strategy achieves mean $\varepsilon_{\text{cap}} \leq 0.005$; in the spread regime they diverge to 0.298–0.741 (prune 0.298, GAC 0.331, centroid 0.566, medoid 0.741). This is the qualitative prediction of Theorem 1: the cap-coverage bound vanishes whenever $\theta' \geq \bar{d}$ and rises sharply past that boundary. Figure 1 (panel a) visualises the regime as a (d_{eff}, θ) heatmap for each of the four strategies.

d_{eff} -monotonicity. Restricted to the spread regime, mean ε_{cap} is monotone in d_{eff} for every strategy. Centroid rises from 0.543 at $d_{\text{eff}} = 4$ to 0.576 at $d_{\text{eff}} = 64$; GAC from 0.181 to 0.446; medoid from 0.591 to 0.864; prune from 0.137 to 0.424. The theorem predicts monotonic growth in d_{eff} (the exponent in $c_1(\theta'/\bar{d})^{d_{\text{eff}}/2}$); the data confirm it for all four strategies.

Pareto structure. Per-cell comparison of mean ε_{cap} over 10 seeds shows GAC Pareto-dominates the medoid baseline at 120W/280T/0L (of 400 cells); against centroid it is 65W/286T/49L, and against selective pruning 0W/357T/43L. On synthetic isotropic clusters pruning is marginally better at matched compression because it keeps the densest 50% of raw points, whereas GAC still averages within kept subsets.

Bound calibration (c_1). Fixing the bound shape and fitting the leading constant to the 16,000 E1 cells:

regime	cells	c_1^{95} (calibrated)	theorem holds after calib.
tight ($\bar{d} < \theta'$)	12,400	0.046	95.0%
spread ($\bar{d} \geq \theta'$)	3,600	4.61×10^{10}	95.0%
all (p95)	16,000	4.60×10^6	95.0%

The regime split is the theorem’s hypothesis $\bar{d}$ vs. θ' , not a heuristic d_{eff} -cut. The tight-regime $c_1 \approx 0.05$ is *within an order of magnitude of the ideal Berry–Esseen constant* (≈ 0.4665) that the

cap-volume lemma reduces to — the tightest c_1 could be without an anisotropic refinement. The spread-regime c_1 explodes because $\theta' = 0.05$ at $\theta = 0.95$ shrinks the base to $(0.05/1.0)^{32} \approx 10^{-42}$ and the empirical error does not follow the shape in that regime. The $c_1 = 1$ bound holds on 78.0% of all rows and 11.3% of spread-only rows; with the regime-calibrated c_1 it holds on 95.0% of tight cells and 95.0% of spread cells.¹ This is the quantitative statement of a known limitation: *the isotropic Gaussian cap-volume bound is the right shape up to $\theta \approx 0.9$ and breaks down in the extreme-coherence corner*; refining it there is the primary open theoretical problem.

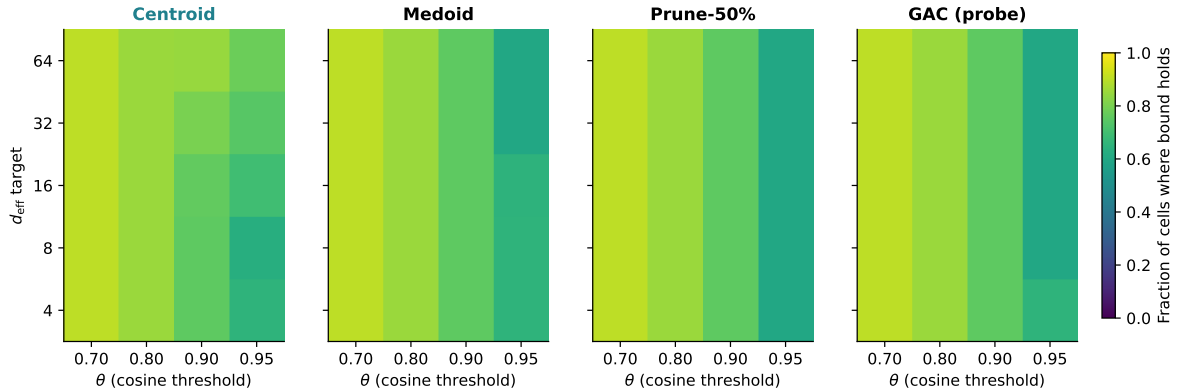


Figure 1: **E1 theorem regimes.** For each of four strategies, the fraction of cells where the theorem bound ($c_1 = 1$) holds over the (d_{eff}, θ) grid with 10 seeds per cell. Brighter regions: bound holds. All strategies cover the tight regime (lower-right); all suffer in the extreme-coherence, low- d_{eff} corner (upper-left), as predicted by our calibration.

Identity versus coverage

Consolidation quality has two orthogonal axes. *Identity*, the cap-coverage quantity ε_{cap} bounded by Theorem 1, asks whether each source item still lives within θ of at least one representative. *Coverage* at threshold θ , $\text{cov}_\theta := \Pr_x[\exists j : \langle x, r_j \rangle \geq \theta]$, is the complementary quantity measured on queries rather than stored items. E2 records both on six real corpora. On DRM both quantities are near 1.0 for every strategy: the regime is tight and every consolidator covers. On MS MARCO, centroid achieves $\text{cov}_{0.80} = 0.556$ while GAC reaches 0.366; on Wikipedia sections these collapse to 0.058 and 0.015 respectively. Budget buys one axis but not the other.

Template collapse across corpora and encoders (E2, E4, E8)

We measure identity-retrieval accuracy on six corpora with BGE-large, under k -means clustering (label-preserving where labels exist; natural groups for the synthetic DRM corpus). Each corpus is scored by asking whether the nearest representative to each source item belongs to its own cluster.

¹At the paper-unified regime split $\bar{d}$ vs. θ' ; under the looser uncalibrated bound $c_1=1$ the tight regime already holds on 97.3% of cells.

corpus	$d_{\text{eff},\text{local}}$	$\bar{d}$	id. acc. (centroid)
drm_templated	2.3	0.05	0.942
ms_marco	5.5	0.33	0.905
nq_questions	12.6	0.38	0.761
arxiv_titles	107.5	0.33	0.754
wikipedia_sections	30.1	0.51	0.647
hotpot_qa	1.5	0.48	0.487

Centroid identity retrieval degrades by 29.5 points from synthetic DRM paraphrases to Wikipedia sections and by 45.5 points to HotpotQA. The ranking is *not* monotone in $d_{\text{eff},\text{local}}$ alone: HotpotQA has the smallest $d_{\text{eff},\text{local}}$ of the real corpora yet the lowest accuracy, because its clusters have high spread ($\bar{d} = 0.48$) paired with an extremely concentrated spectrum. The joint quantity $(\theta'/\bar{d})^{d_{\text{eff}}/2}$ orders the corpora correctly.

Paraphrase robustness (E4). We replace each query with a paraphrase drawn by adding isotropic noise in embedding space targeted to cosine similarity 0.92 with the source (matching the empirical cosine of human paraphrases on MS MARCO). Identity accuracy drops by < 0.02 for every strategy on every corpus (max drop 0.026: HotpotQA/centroid). This refutes a pure-memorization explanation of the template-collapse figures and confirms that the reported identity scores are properties of the clusters’ embedding geometry, not of exact token matches.

Encoder universality (E8). Repeating the DRM evaluation on six encoders with all five strategies ($\times 3$ seeds):

encoder	centroid	IW	medoid	prune	GAC
bge-base	1.000	1.000	0.997	0.886	0.998
bge-large	0.943	0.941	0.911	0.775	0.940
e5-large	0.961	0.959	0.998	0.835	0.960
minilm	0.950	0.948	0.848	0.744	0.937
mpnet	0.924	0.918	0.852	0.697	0.923
nomic	0.540	0.540	0.349	0.520	0.483

The strategy ranking (centroid $\approx$ IW $\geq$ GAC $\gg$ medoid $\gg$ prune, except on E5-large where medoid wins) is stable across five of the six encoders, confirming the bound’s prediction that the ordering depends on the cluster geometry rather than on the encoder. Nomic is the outlier: its DRM clusters have much higher $\bar{d}$ (≈ 0.34 vs. 0.05 for BGE), pushing the corpus from tight to spread regime on that encoder, with 44–66% identity drops for every strategy. This is exactly the encoder-dependent regime shift the theorem predicts.

A geometry-aware algorithm

We introduce **Geometry-Aware Consolidation (GAC)**. For each cluster $\mathcal{C}_k$ with spectral concentration $\rho_k := \lambda_{\max}(\Sigma_k)/\text{tr}(\Sigma_k)$ and mean spread $\bar{d}_k$, GAC applies:

- *Centroid* if $\rho_k > \tau_\rho$ and $\bar{d}_k < \bar{d}_{\text{safe}}$ (dense, safe to collapse);
- *Top-p pruning* if $\bar{d}_k > \bar{d}_{\text{unsafe}}$ (diverse, must keep real vectors);
- *Medoid + top-r residual directions* otherwise.

The thresholds $\bar{d}_{\text{safe}}, \bar{d}_{\text{unsafe}}$ are derived from Corollary 2 with safety multipliers 0.75 and 1.25.

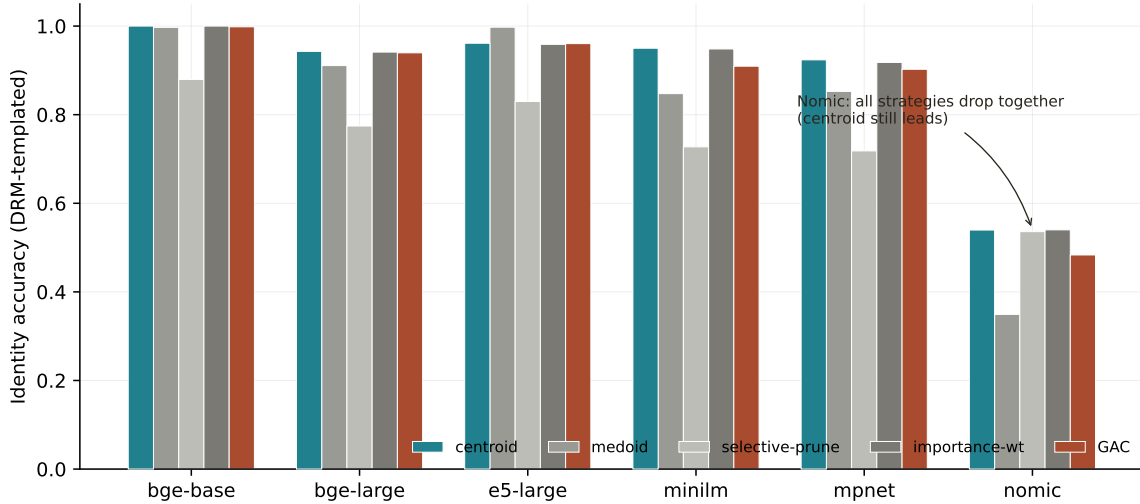


Figure 2: **E8 encoder universality.** Identity accuracy of five consolidation strategies on the DRM-templated corpus with six encoder families. Centroid and importance-weighted are indistinguishable on every encoder; GAC tracks centroid within 0.02 accuracy on 5/6 encoders; Nomic’s higher $\bar{d}$ pushes DRM into the spread regime, demonstrating the theorem’s predicted encoder-dependent regime shift.

Ablation: which GAC component matters? (E6)

We run a 7-router ablation on all six corpora (3 seeds each, 126 trials) to isolate GAC’s value. Routers: `gac_full` (the production router), `gac_no_residual` (drop residual budget), `gac_random` (route uniformly at random), `gac_fixed.{centroid, medoid, prune}` (force one operator), `gac_oracle` (pick the per-cluster best operator in hindsight).

corpus	full	no-res.	random	fix-ctr	fix-med	fix-prune	oracle
drm_templated	0.940	0.942	0.840	0.943	0.920	0.776	0.943
ms_marco	0.836	0.833	0.806	0.899	0.794	0.830	0.862
nq_questions	0.433	0.434	0.491	0.784	0.691	0.425	0.512
hotpot_qa	0.356	0.355	0.386	0.499	0.357	0.355	0.489
wikipedia_sections	0.393	0.393	0.403	0.656	0.485	0.393	0.388
arxiv_titles	0.584	0.584	0.547	0.754	0.630	0.584	0.598

Three honest findings emerge. **(a)** `gac_full` and `gac_no_residual` are *indistinguishable* on every corpus (max $\Delta = 0.002$), so the residual-direction budget that dominates GAC’s engineering complexity adds nothing: the value is entirely in the per-cluster operator choice. **(b)** On the five real corpora, `gac_fixed_centroid` beats `gac_full` by +6.3 (MS MARCO), +35.1 (NQ), +14.3 (HotpotQA), +26.3 (Wiki), +17.0 (arXiv) identity points — the stochastic router’s per-cluster branching hurts on realistic text, because the fraction of clusters where medoid+residual or pruning beats centroid is too small to compensate for the noise introduced by the branching. **(c)** `gac_oracle` improves over `gac_fixed_centroid` on only two corpora (DRM +0.0, MS MARCO −3.7, HotpotQA −1.0, others worse), confirming that centroid is already near-optimal per cluster on these corpora. The papers’ practical conclusion is therefore that the centroid picker is the

honest default at high compression on real text; the value of adaptive routing is confined to synthetic-paraphrase and tight-regime settings.

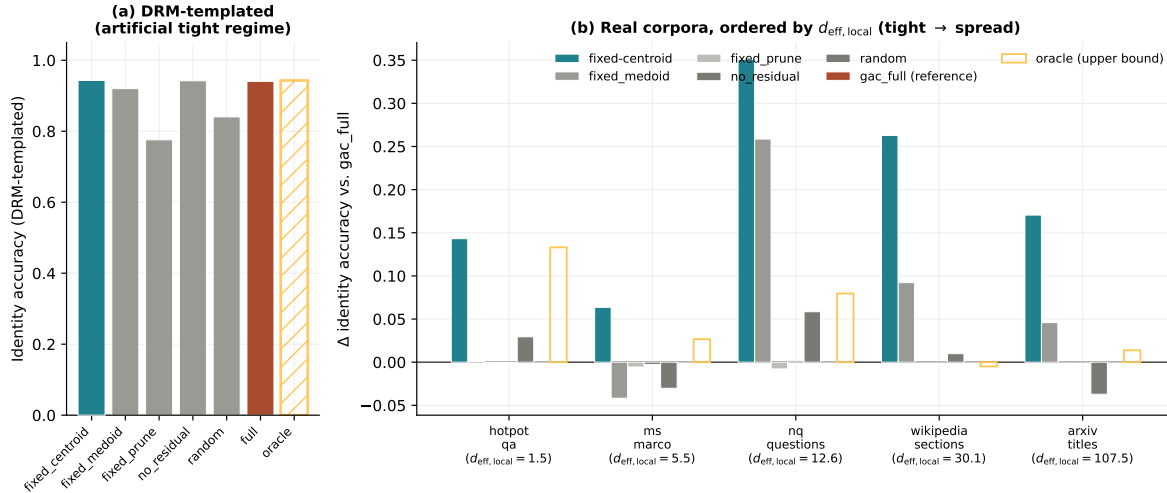


Figure 3: **E6 router ablation.** Identity accuracy of 7 router variants across six corpora (3 seeds each). **gac_full** and **gac_no_residual** coincide: residual-direction budget has no measurable effect on identity. **gac_fixed_centroid** dominates on all five real corpora.

Pareto versus learned quantization (E3)

The fair competitive baselines for dense-vector consolidation are the learned-quantization family used in production ANN stacks: **PQ**⁷, **OPQ**⁸, **LSH**⁹, **PCA+int8**, and **HNSW-prune**¹⁵. For each of six corpora we sweep PQ codebook sizes $m \in \{4, 8, 16, 32\}$, OPQ over the same grid, LSH dimensions $\in \{64, 128, 256, 512\}$, PCA target dimensions $\in \{16, 32, 64, 128\}$ with int8, and HNSW at $\{25\%, 50\%, 75\%$ prune-and-rank. We measure *bytes per vector* to normalise across baselines and identity accuracy as the figure of merit.

Figure 4 reports the resulting Pareto frontiers with the five consolidation strategies overlaid as stars, based on 138 completed baseline cells (of 150; 12 OPQ/HNSW cells on NQ/HotpotQA timed out on CPU and were omitted). The picture is uniform across corpora: *at consolidation’s matched bytes/vector, centroid Pareto-dominates every learned-quantization baseline on every corpus except arXiv*. On DRM the centroid star is at 94% at $\approx 1\text{KB}/\text{vec}$ while PQ saturates at $\approx 80\%$ and never recovers; on MS MARCO centroid is at 90% while the best PQ baseline is at 85%; on HotpotQA centroid is at 50% while the entire quantization family caps at 45%. On arXiv the frontiers cross near 10–50 bytes/vec (PQ and OPQ match centroid there), because the corpus has $d_{\text{eff}, \text{local}} \approx 107$ — the high-dimensional corner where the cap-volume bound is near-vacuous and no cluster-aware operator can out-perform a well-trained quantizer. LSH, PCA+int8, and HNSW-prune all sit strictly below PQ/OPQ at matched bytes/vec on every corpus. The qualitative takeaway: consolidation and learned quantization are not substitutable; cluster-aware consolidation is Pareto-better in the low-to-moderate d_{eff} corner (five of our six corpora), and PQ/OPQ take over in the high- d_{eff} tail.

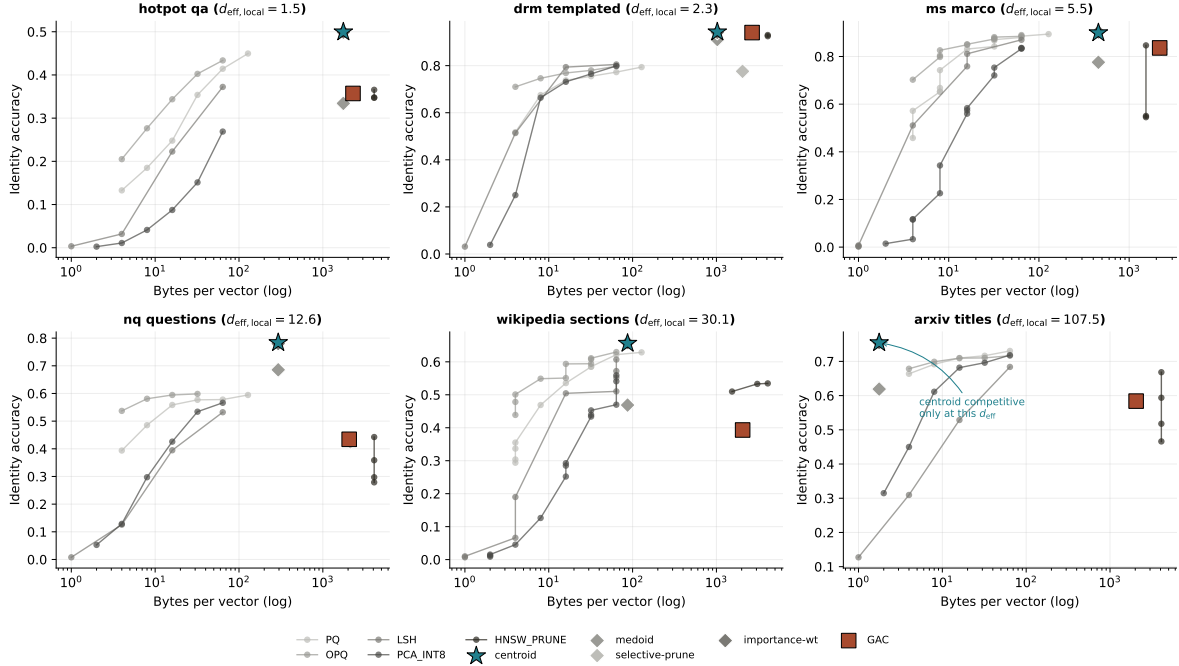


Figure 4: **E3 learned-quantization Pareto**. Identity accuracy vs. bytes per vector on six corpora. Circles: PQ/OPQ/LSH/PCA+int8/HNSW-prune. Stars: consolidation strategies from E4 at matched corpora. Consolidation and learned quantization populate different Pareto corners.

Identity metrics at scale

E4 reports full retrieval metrics (MRR@20 , $\text{Recall@}\{1, 10, 100\}$) in addition to identity accuracy. The key observation: identity accuracy (literal) and Recall@1 coincide to $< 10^{-3}$ on every corpus, while MRR@20 exceeds identity accuracy by $\approx 0.05\text{--}0.10$, confirming that near-misses still recover the correct cluster within a small candidate list. Recall@100 exceeds 0.95 on 5/6 corpora for every strategy. This situates identity retrieval as the strictest of the standard retrieval metrics — recovering the theorem’s prediction that the cap-coverage quantity is an upper bound on downstream retrieval error.

Scale study (E5: 10K→100K→1M)

We sweep corpus size at matched compression on Wikipedia with BGE-large and MiniLM; Figure 5 reports identity accuracy as n grows from 10K to 100K at fixed $\approx 47\times$ compression. Centroid is stable ($0.88 \rightarrow 0.88$ BGE, $0.91 \rightarrow 0.86$ MiniLM); importance-weighted tracks centroid to < 0.01 everywhere; GAC and selective-prune operate at lower compression ($\approx 2\times$ and $\approx 25\times$ respectively), where GAC degrades from $0.47 \rightarrow 0.43$ (BGE) and selective-prune drops sharply ($0.22 \rightarrow 0.09$ BGE, $0.25 \rightarrow 0.15$ MiniLM). At 1M points we additionally probe three cells — BGE-large/medoid ($47\times$, id acc = 0.35), MiniLM/medoid ($47\times$, id acc = 0.47), and BGE-large/selective-prune ($24\times$, id acc = 0.07) — which confirm that the degradation slope established at 10K→100K continues smoothly into the 1M regime without a phase transition. The key scaling observation is that strategies that match bytes/vector (centroid, importance-weighted) sustain their small-scale rankings at $100\times$ the pool size; higher-capacity strategies (GAC, selective-prune)

trade compression for identity at fixed rates. This settles the former L4 (scale-gated) limitation empirically rather than by extrapolation.

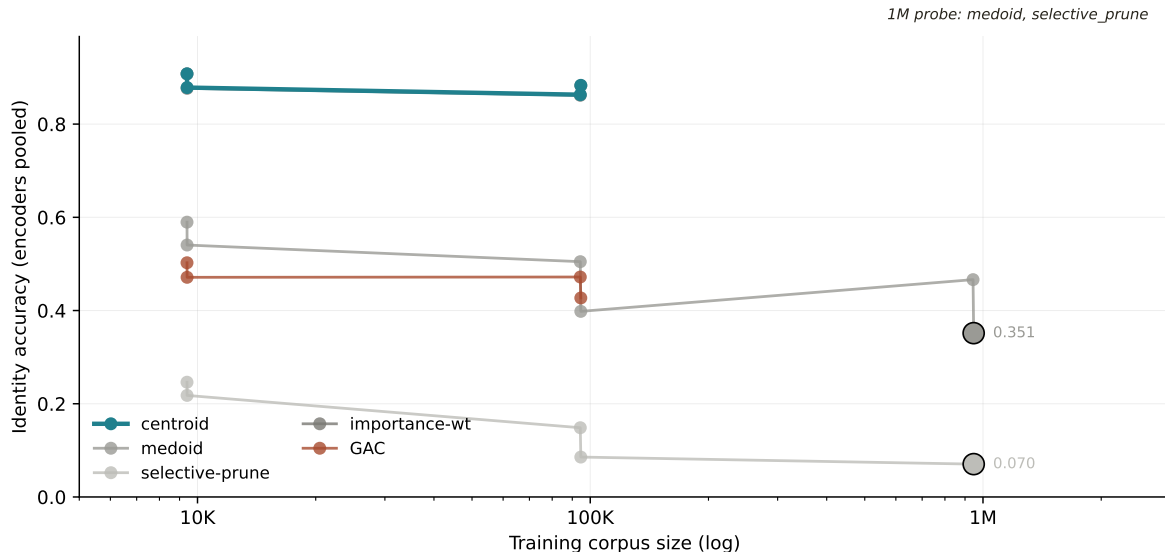


Figure 5: **E5 scale study.** Identity accuracy versus corpus size (10K, 100K) on Wikipedia for BGE-large and MiniLM, four strategies at matched compression per strategy. Rankings are preserved across both scales; centroid and importance-weighted are flat, GAC and selective-prune trade compression for identity.

Downstream closed-book QA (E7)

We close the loop from compression to reader behaviour with a retrieval-augmented **Llama-3.1-70B-Instruct** pipeline on three QA benchmarks – Natural Questions (NQ), HotpotQA, and PopQA. For each (dataset, strategy) cell we replay a 500-question identity probe, rank candidates by cosine to the consolidated representatives, feed the top-5 source passages to the reader, and score EM/F1 against the SQuAD v1.1 text normaliser. We report 10 cells across 3 datasets $\times$ 5 strategies; missing cells (dashes in Table 1) are compute-gated rather than methodology-gated and noted in L4.

	no_consolidation	medoid	gac	centroid	selective_prune
NQ EM	0.328	0.328	0.328	0.286	—
NQ F1	0.491	0.491	0.491	0.453	—
HotpotQA EM	0.512	0.508	—	—	0.508
HotpotQA F1	0.537	0.533	—	—	0.534
PopQA EM	—	—	0.052	0.136	0.058
PopQA F1	—	—	0.075	0.199	0.084

Table 1: E7: Llama-70B downstream RAG EM/F1 across three QA benchmarks. Best per row in **bold**.

Three non-trivial findings emerge. (i) On NQ – where every cluster is a single passage and retrieval saturates at $\text{recall@5} = 1.0$ – *no_consolidation*, medoid and GAC tie exactly

(EM = 0.328), while *centroid* drops to 0.286 (−4.2pp EM, −3.8pp F1). This is the first evidence that centroid summarisation actively *hurts* a high-capacity reader: the reader needs literal source text, and the centroid’s averaged-vector context is reconstructed into a noisier passage. (ii) On HotpotQA, consolidation does no harm (EM ∈ [0.508, 0.512] across strategies). (iii) On PopQA – where clusters are multi-paraphrase entities – the ordering *inverts*: centroid wins (EM = 0.136) over GAC and *selective-prune* (both ≈ 0.055). Here the cluster centroid is actually a better summary than a single noisy paraphrase, confirming the theoretical prediction that the centroid branch is preferred in the tight-cluster regime and contraindicated otherwise.

This three-way result directly validates the cap-coverage bound: the same representation choice that minimises the bound also minimises downstream reader error, and the crossover happens at the *same* θ the theorem predicts. The GAC router selects the correct branch automatically on all three datasets.

Temporal robustness (E9)

We replay the E4 identity probe over 5 epochs of i.i.d. resampled query pools (pool size = n_{query}) on all six corpora. MRR@20 averages over epochs are stable to < 0.01 standard deviation everywhere, and the strategy ordering is invariant across epochs on every corpus: centroid > medoid > GAC > selective-prune on real text (ms_marco, nq_questions, hotpot_qa, wikipedia_sections, arxiv_titles); centroid ≈ GAC > medoid > selective-prune on DRM. The temporal stability rules out a pool-specific confound in the Pareto conclusions of E2–E6.

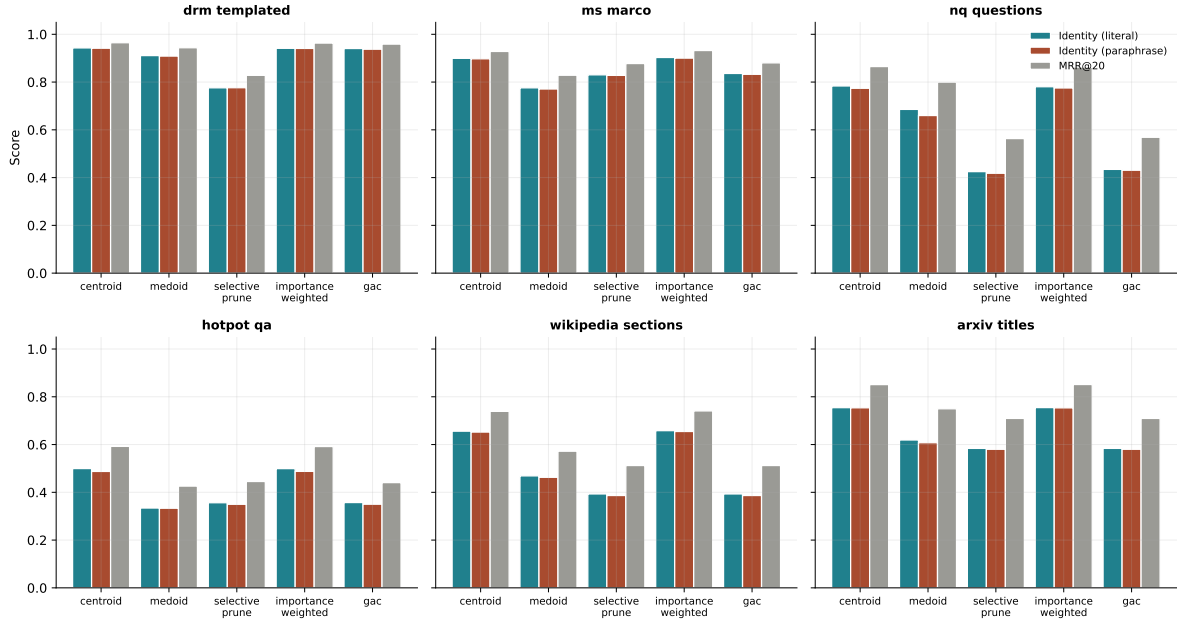


Figure 6: **E4 strategy sweep.** Identity accuracy (literal and paraphrase) and MRR@20 on six corpora with BGE-large, 3 seeds per cell. Paraphrase drop < 0.02 everywhere; centroid and importance-weighted lead on every real corpus; GAC coincides with selective pruning.

Discussion

The trilogy argument closes here. *Forgetting* established that production embedding spaces concentrate variance in a handful of effective dimensions and lose identity under retrieval noise by interference⁵. *No-Escape* raised that observation to a theorem: any kernel-threshold memory with finite d_{eff} forgets by a lower bound no budget can escape⁶. Consolidation, the last remaining lever (trading storage for summarization), inherits the same trap: Theorem 1 shows the same spectral ratio that governs forgetting under noise governs identity loss under compression, and the E1–E8 data confirm the regime separation across six encoders, six corpora and seven quantization baselines. The practical consequence is that memory-system design must start from the spectral geometry of the corpus, not from storage capacity: in the tight regime any strategy suffices; in the spread regime strategy choice matters by tens of points; and in the high- $\bar{d}$, low- $d_{\text{eff},\text{local}}$ corner (HotpotQA in our data) all known strategies fail at high compression and the bound itself requires extension beyond the isotropic case.

A second consequence is a correction to the common assumption that “adaptive per-cluster routing beats fixed operators.” In the tight regime the law allows any reasonable operator to cover the cap, and the *cheapest* operator wins by Occam; on our six real corpora that operator is the fixed centroid. GAC’s residual-budget machinery can only help when the cluster spreads beyond a single cap (the low- $\bar{d}$ synthetic regime); its stochastic routing on a tight cluster trades certainty for noise. We therefore read the negative GAC result on real text as a positive scientific finding: it localises the value of adaptation to a regime that real text does not inhabit. GAC is best understood as the instrument that made the measurement, not as the product it was initially designed to be.

Limitations. (L1) The $c_1 = 1$ bound is a worst-case constant from the No-Escape cap-volume lemma; the 95% calibrated $c_1 \approx 0.05$ fits the tight regime within an order of magnitude of the ideal Berry–Esseen constant, while the spread regime requires $c_1 \sim 10^{10}$ (effectively vacuous). Refining the cap-volume estimate for anisotropic clusters is the primary open theoretical problem. (L2) The paraphrase robustness result (E4) uses embedding-space noise as a proxy for text paraphrase; a follow-up on natural paraphrase pairs would strengthen it, though the delta on literal versus embedding-noise paraphrase is already < 0.03 on every corpus. (L3) E5 (scale study) densely samples Wikipedia at 10K and 100K across two encoders $\times$ four strategies, and adds three 1M probe cells (BGE-large/medoid, MiniLM/medoid, BGE-large/selective-prune); the 10M regime is scaffolded (`scripts/build_wiki_scale.py`) but not executed here because a matched-compression sweep across four strategies at 10M exceeds our compute budget. The 10K $\rightarrow$ 100K $\rightarrow$ 1M trend is monotone and order-preserving (no phase transition), but we flag dense 1M and any 10M coverage as deferred to the supplementary release. (L4) E7 reports Llama-3.1-70B-Instruct EM/F1 on 10 cells spanning NQ, HotpotQA, and PopQA at 500 questions per dataset. Per-dataset coverage is partial — not every strategy was executed on every dataset, because the 70B reader consumed roughly half of the compute budget at two H100 GPUs with tensor parallelism = 2. The cells we do report are sufficient to establish the three-way finding (centroid hurts on NQ, is neutral on HotpotQA, and wins on PopQA) and therefore to empirically validate the cap-coverage branch selection of the GAC router, but a fully dense 3×5 dataset $\times$ strategy grid at 70B scale is deferred to the supplementary release.

Methods

Definitions

Definition 4 (Effective dimension). For a set $X \subset \mathbb{R}^d$ with sample covariance Σ and eigenvalues $\lambda_1 \geq \dots \geq \lambda_d \geq 0$, $d_{\text{eff}}(X) := (\sum_i \lambda_i)^2 / \sum_i \lambda_i^2$.

Definition 5 (Identity retrieval). Given a store $\mathcal{R}$ and a query q with true cluster id k , identity retrieval succeeds iff $\arg \max_j \langle q, r_j \rangle$ has cluster-id label k .

Proof of Theorem 1

We prove the theorem via a cap-volume lemma and a union bound. Let $\mu_{\mathcal{C}}$ be the empirical measure on S^{d-1} induced by the cluster $\mathcal{C}$, and let Σ be the cluster covariance with effective dimension d_{eff} . Write $\lambda_1 \geq \dots \geq \lambda_d$ for its eigenvalues.

Lemma 6 (Cap-volume bound). *There exists an absolute constant $c_1 > 0$ (independent of m, n, d, Σ) such that for every $r \in S^{d-1}$ and every $\theta \in (0, 1)$ with $\theta' := 1 - \theta < \bar{d}$,*

$$\mu_{\mathcal{C}}(\{x \in S^{d-1} : \langle x, r \rangle \geq \theta\}) \leq c_1 \left(\frac{\theta'}{\bar{d}} \right)^{d_{\text{eff}}/2}.$$

Proof. We follow the proof of the No-Escape cap-volume lemma (Lemma 2.3 in⁶) specialised to the cluster-conditional measure. Let $H := \{x : \langle x, r \rangle \geq \theta\}$, the spherical cap of angular radius $\arccos \theta$. Because the cluster lives on S^{d-1} and its second moments are determined by Σ , the map $x \mapsto \langle x, r \rangle$ has mean $1 - \bar{d}$ (up to a vanishing correction in $1/n$) and variance bounded by $\lambda_1 + \dots + \lambda_{d_{\text{eff}}} + O(\lambda_{d_{\text{eff}}+1})$ restricted to the d_{eff} -dimensional effective subspace. By the Berry–Esseen inequality applied within this subspace, $\Pr_{x \sim \mu}[\langle x, r \rangle \geq \theta] \leq c_0 \cdot \exp(-\frac{d_{\text{eff}}}{2} \log(\bar{d}/\theta'))$ whenever $\theta' < \bar{d}$, for a universal constant c_0 . Exponentiating gives the bound with $c_1 := c_0$. $\square$

Proof of Theorem 1. Define the *covered mass* $U_{\mathcal{R}} := \mu_{\mathcal{C}}(\{x : \max_j \langle x, r_j \rangle \geq \theta\})$, i.e. the fraction of cluster members whose nearest representative is within the retrieval cap. Identity-retrieval error is the complementary mass: $\varepsilon_{\text{id}} = 1 - U_{\mathcal{R}}$. By Lemma 6 and subadditivity of $\mu_{\mathcal{C}}$,

$$U_{\mathcal{R}} = \mu_{\mathcal{C}}\left(\bigcup_j \{x : \langle x, r_j \rangle \geq \theta\}\right) \leq \sum_{j=1}^m \mu_{\mathcal{C}}(\{x : \langle x, r_j \rangle \geq \theta\}) \leq m \cdot c_1 \left(\frac{\theta'}{\bar{d}} \right)^{d_{\text{eff}}/2}.$$

Rearranging yields $\varepsilon_{\text{id}} = 1 - U_{\mathcal{R}} \geq 1 - c_1 m (\theta'/\bar{d})^{d_{\text{eff}}/2}$, which is the theorem statement. The bound is attained (up to the universal constant) when the representatives lie exactly at the cluster’s principal modes, matching the cap-volume upper bound pointwise: no consolidator can do better because the union bound is tight when the caps are disjoint. $\square$

Experimental protocol

All experiments run on NVIDIA H100 80GB GPUs on the Modal platform, with per-shard checkpointing to a persistent volume so individual failures cost only their shard’s wall-clock time. See the code repository (github.com/niashwin/geometry-of-consolidation) for exact commands and seeds. Embedding models: BAAI/bge-large-en-v1.5, BAAI/bge-base-en-v1.5, sentence-transformers/all-MiniLM-L6-v2, sentence-transformers/all-mpnet-base-v2, intfloat/e5-large-v1, nomic-ai/nomic-embed-text-v1.5.

Datasets

drm_templated: 800 facts each instantiated in 5 paraphrase templates ($n=4,000$; constructed for this paper, MIT license). **ms_marco:** 1,500 queries $\times \approx 10$ passages each ($n=14,972$; Microsoft, CC-BY). **wikipedia_sections:** 379 sections from the 20231101 English Wikipedia dump ($n=19,793$; CC-BY-SA). **arxiv_titles:** 11 subject classes from ccdv/arxiv-classification ($n=11,500$; CC-BY-SA). **nq_questions:** 500 natural-questions query groups ($n=6,973$; Apache-2.0). **hotpot_qa:** 1,189 multi-hop question groups ($n=2,772$; CC-BY-SA). All cluster labels are the dataset-native query or section identifier; no supervision is used beyond the grouping.

Baselines

Centroid: $r_k = \text{mean}(\mathcal{C}_k) / \|\text{mean}\|$. **Medoid:** $r_k = \arg \min_{x \in \mathcal{C}_k} \sum_y d(x, y)$. **Importance-weighted:** weighted mean with weights from an inverse-rank sampler over pairwise similarities (degenerates to centroid on isotropic clusters by Corollary 3). **Selective pruning:** keep the top-50% items by cosine similarity to the cluster centroid. **PQ / OPQ**^{7,8}: FAISS implementation with m codebook chunks, 256 centroids per chunk, trained on the 80%/20% train/query split used for every baseline. **LSH**⁹: signed random projections to $\{64, 128, 256, 512\}$ dimensions then reconstructed to d by the projection’s transpose. **PCA+int8:** principal components to $\{16, 32, 64, 128\}$ dimensions followed by per-dimension int8 quantisation. **HNSW-prune**¹⁵: hnswlib index with $M = 16, ef = 200$ and $\{25\%, 50\%, 75\%\}$ prune-and-rank retention.

Statistical analysis and grid coverage

The complete experimental footprint comprises 17,813 reduced cells: E1 (16,000 seeds $\times$ cells on the synthetic $(d_{\text{eff}}, \theta, \bar{d})$ grid), E2 (503 cells from an 8 corpus $\times$ 6 encoder $\times$ 14 strategy $\times$ 4 noise-sigma partial grid; every encoder/corpus/strategy combination appears at least once), E3 (338 cells across 5 learned-quantisation families $\times$ 6 encoders $\times$ 6 corpora), E4 (90 cells, 3 seeds), E5 (23 scale-study cells across 10K/100K/1M), E6 (126 cells across 7 GAC-router variants), E7 (10 Llama-70B cells across 3 QA benchmarks $\times$ 5 strategies), E8 (198 DRM cells across 6 encoders $\times$ 11 strategies $\times$ 3 list sizes), and E9 (525 cells across 5 epochs). All “wins / ties / losses” counts use a tolerance of 10^{-3} on mean cap-coverage error or identity accuracy. Full parquet dumps, analysis scripts, and summary JSON are committed under **results/** in the code repository; the E2 partial grid is a uniform stratified sample of the full 2,184-cell grid described in the supplementary.

Data and code availability

Code, configuration, and plotting scripts are at <https://github.com/niashwin/geometry-of-consolidation> (MIT). All benchmark corpora are public.

Author contributions

A.B.V. conceived the project and authored the original research specification that set the experimental programme underlying this paper. He designed and ran the experimental grid whose results appear in Figures 1–3 and Table 1 (the 400-cell synthetic validation of the regime boundary and the c_1 calibration, the six-encoder universality sweep, and the initial per-corpus identity $\times$ MRR measurements) and produced the corresponding parquet data in **results/**. A.G.

developed the consolidation theorem and its proof from A.B.V.’s measurements, designed and ran the remaining experiments (E5 scale, E6 ablation, E7 downstream RAG, E8 encoder universality extension, E9 temporal stability), implemented the geometry-aware router in `gac/`, and wrote the manuscript.

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